ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 1.01: Functions, Modulus & Quadratics

Q1. Solve the modulus equation |2x + 3| = 7 .

[ ] A: x = 3 and x = -3

[ ] B: x = -2 and x = -frac{10}{2}

[ ] C: x = 2 only

[x] D: x = 2 and x = -frac{10}{2}

Solution: Case 1: 2x + 3 = 7 implies 2x = 4 implies x = 2 . Case 2: 2x + 3 = -7 implies 2x = -10 implies x = -frac{10}{2} .

Q2. Solve the modulus equation |3x + 3| = 9 .

[x] A: x = 2 and x = -frac{12}{3}

[ ] B: x = 3 and x = -3

[ ] C: x = -2 and x = -frac{12}{3}

[ ] D: x = 2 only

Solution: Case 1: 3x + 3 = 9 implies 3x = 6 implies x = 2 . Case 2: 3x + 3 = -9 implies 3x = -12 implies x = -frac{12}{3} .

Q3. Solve the modulus equation |4x + 3| = 11 .

[ ] A: x = 2 only

[ ] B: x = 3 and x = -3

[x] C: x = 2 and x = -frac{14}{4}

[ ] D: x = -2 and x = -frac{14}{4}

Solution: Case 1: 4x + 3 = 11 implies 4x = 8 implies x = 2 . Case 2: 4x + 3 = -11 implies 4x = -14 implies x = -frac{14}{4} .

Q4. Solve the modulus equation |5x + 3| = 13 .

[x] A: x = 2 and x = -frac{16}{5}

[ ] B: x = 2 only

[ ] C: x = -2 and x = -frac{16}{5}

[ ] D: x = 3 and x = -3

Solution: Case 1: 5x + 3 = 13 implies 5x = 10 implies x = 2 . Case 2: 5x + 3 = -13 implies 5x = -16 implies x = -frac{16}{5} .

Q5. Solve the modulus equation |6x + 3| = 15 .

[ ] A: x = 3 and x = -3

[x] B: x = 2 and x = -frac{18}{6}

[ ] C: x = 2 only

[ ] D: x = -2 and x = -frac{18}{6}

Solution: Case 1: 6x + 3 = 15 implies 6x = 12 implies x = 2 . Case 2: 6x + 3 = -15 implies 6x = -18 implies x = -frac{18}{6} .

Q6. Solve the modulus equation |7x + 3| = 17 .

[x] A: x = 2 and x = -frac{20}{7}

[ ] B: x = 2 only

[ ] C: x = -2 and x = -frac{20}{7}

[ ] D: x = 3 and x = -3

Solution: Case 1: 7x + 3 = 17 implies 7x = 14 implies x = 2 . Case 2: 7x + 3 = -17 implies 7x = -20 implies x = -frac{20}{7} .

Q7. Solve the modulus equation |8x + 3| = 19 .

[x] A: x = 2 and x = -frac{22}{8}

[ ] B: x = 2 only

[ ] C: x = -2 and x = -frac{22}{8}

[ ] D: x = 3 and x = -3

Solution: Case 1: 8x + 3 = 19 implies 8x = 16 implies x = 2 . Case 2: 8x + 3 = -19 implies 8x = -22 implies x = -frac{22}{8} .

Q8. Solve the modulus equation |9x + 3| = 21 .

[ ] A: x = 2 only

[x] B: x = 2 and x = -frac{24}{9}

[ ] C: x = 3 and x = -3

[ ] D: x = -2 and x = -frac{24}{9}

Solution: Case 1: 9x + 3 = 21 implies 9x = 18 implies x = 2 . Case 2: 9x + 3 = -21 implies 9x = -24 implies x = -frac{24}{9} .